

TASK 7:Sum of even or odd digits

```
import java.util.Scanner;

public class DigitSum {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        long number = scanner.nextLong();

        // Handle negative numbers by taking the absolute value
        long temp = Math.abs(number);

        int evenSum = 0;
        int oddSum = 0;

        // Loop to extract and process each digit
        while (temp > 0) {
            int digit = (int) (temp % 10); // Extract the last digit

            if (digit % 2 == 0) {
                evenSum += digit;        // Add to even sum
            } else {
                oddSum += digit;         // Add to odd sum
            }

            temp /= 10;                  // Remove the last digit
        }

        // Print the final calculations
        System.out.println("Original Number: " + number);
        System.out.println("Sum of Even Digits: " + evenSum);
        System.out.println("Sum of Odd Digits: " + oddSum);

        scanner.close();
    }
}
```

TASK 8:Nth Fibonacci

```
public class Fibonacci {
    public static int findNthFibonacci(int n) {
        if (n <= 0) return 0;
        if (n == 1) return 1;

        int prev2 = 0; // Represents F(n-2)
```

```

int prev1 = 1; // Represents F(n-1)
int current = 0;

for (int i = 2; i <= n; i++) {
    current = prev1 + prev2;
    prev2 = prev1;
    prev1 = current;
}

return current;
}

public static void main(String[] args) {
    int n = 9;
    System.out.println("The " + n + "th Fibonacci number is: " + findNthFibonacci(n));
}
}

```

TASK 9: Is Palindrome Number

```

import java.util.Scanner;

public class PalindromeExample {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int number = scanner.nextInt();

        if (isPalindrome(number)) {
            System.out.println(number + " is a palindrome number.");
        } else {
            System.out.println(number + " is not a palindrome number.");
        }
        scanner.close();
    }

    public static boolean isPalindrome(int num) {
        // Negative numbers are not palindromes (e.g., -121 reversed is 121-)
        if (num < 0) {
            return false;
        }

        int originalNum = num;
        int reversedNum = 0;

        while (num != 0) {
            int lastDigit = num % 10;        // Extract the last digit

```

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        reversedNum = (reversedNum * 10) + lastDigit; // Build the reversed number
        num = num / 10; // Remove the last digit
    }

    // Compare the reversed number with the original
    return originalNum == reversedNum;
}
}

```

TASK 10: Sum of last digit of two given numbers

```

import java.util.Scanner;

public class SumOfLastDigits {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Input two numbers
        System.out.print("Enter first number: ");
        int num1 = scanner.nextInt();

        System.out.print("Enter second number: ");
        int num2 = scanner.nextInt();

        // Calculate sum of last digits
        int totalSum = addLastDigits(num1, num2);

        // Output result
        System.out.println("The sum of the last digits is: " + totalSum);

        scanner.close();
    }

    /**
     * Extracts the last digit of both numbers and sums them up.
     * Uses Math.abs to properly handle negative integers.
     */
    public static int addLastDigits(int input1, int input2) {
        // Use Math.abs to handle negative numbers correctly
        int lastDigit1 = Math.abs(input1) % 10;
        int lastDigit2 = Math.abs(input2) % 10;

        return lastDigit1 + lastDigit2;
    }
}

```